

UNIT-1



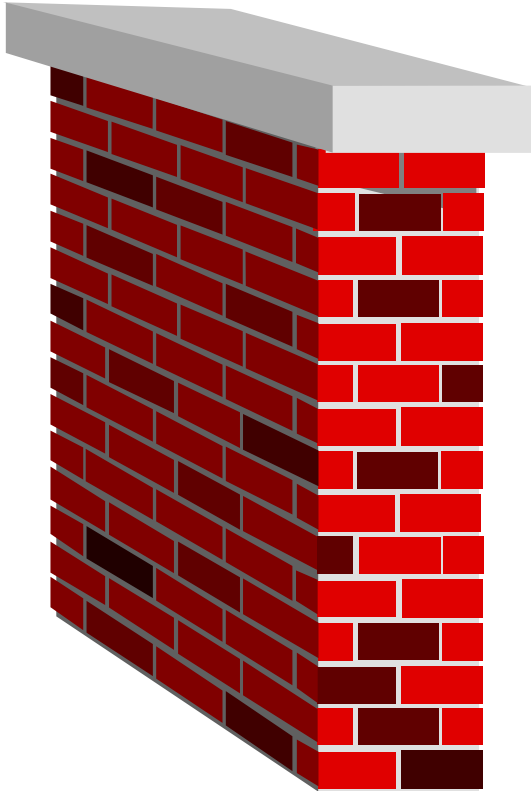
Introduction to Software Engineering

What is Software Engineering?



- Engineering approach to develop software.
- Using Systematic collection of
*past experiences:
techniques,
methodologies,
guidelines.*

Building Construction Analogy.



Software Crisis

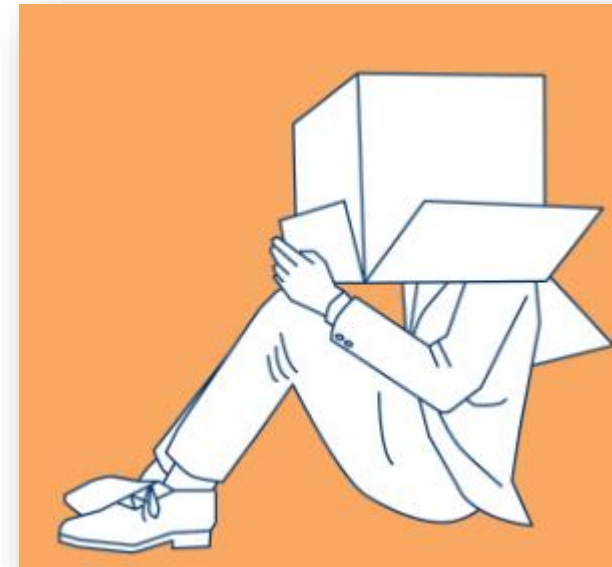
Software products:

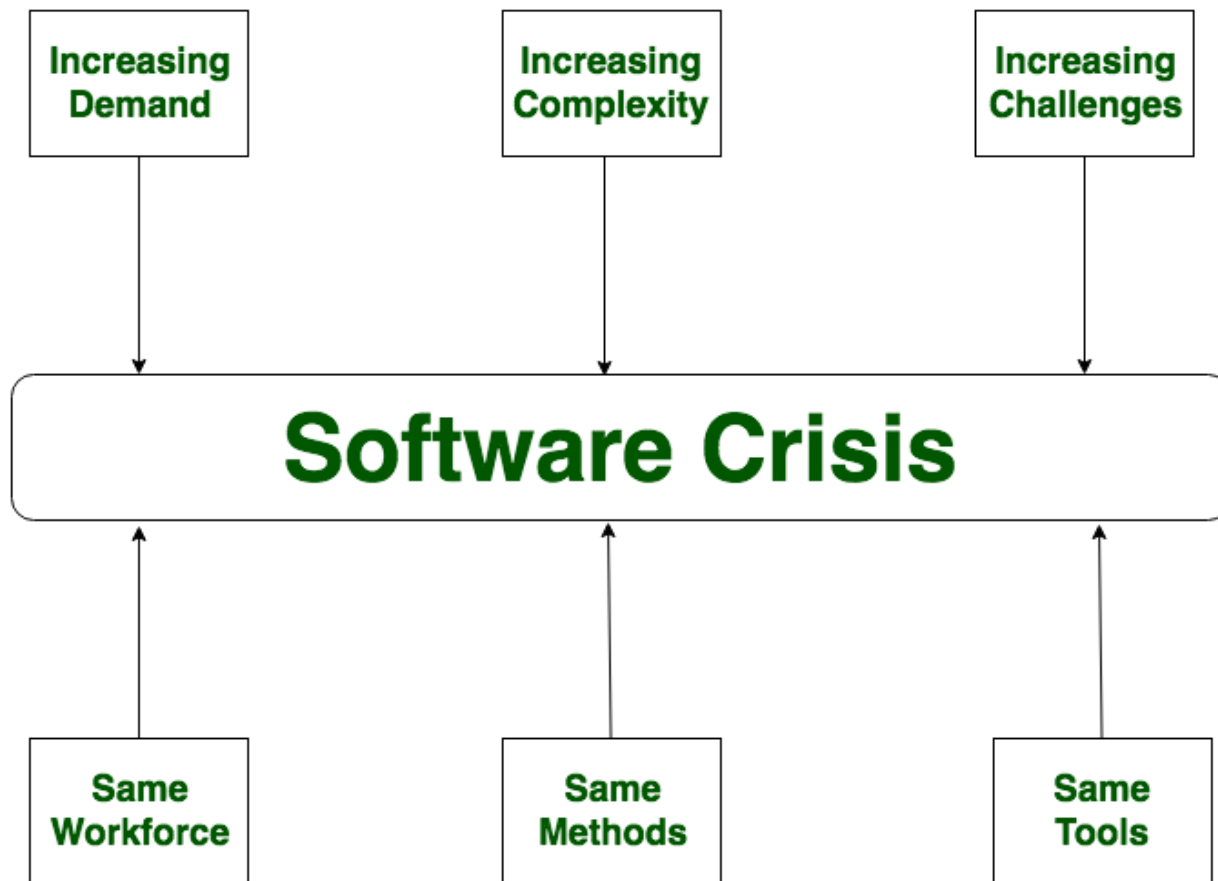
- fail to meet user requirements.
- frequently crash.
- expensive.
- difficult to alter, debug, and enhance.
- often delivered late.
- use resources non-optimally.



Factors contributing to the software crisis

- Larger problems,
- Lack of adequate training in software engineering,
- Increasing skill shortage,
- Low productivity improvements.





Programs versus Software Products

```
1  /*
2  * This line basically imports the "stdio" header file, part of
3  * the standard library. It provides input and output functionality
4  * to the program.
5  */
6  #include <stdio.h>
7
8  /*
9  * Function (method) declaration. This outputs "Hello, world\n" to
10 * standard output when invoked.
11 */
12 void sayHello(void) {
13     // printf() in C outputs the specified text (with optional
14     // formatting options) when invoked.
15     printf("Hello, world!\n");
16 }
17
18 /*
19 * This is a "main function". The compiled program will run the code
20 * defined here.
21 */
22 int main(void)
23 {
24     // Invoke the sayHello() function.
25     sayHello();
26     return 0;
27 }
```

```
28
29
30 // Output:
31 Hello, world!
32
33 // Invoke the sayHello() function.
34 {
35     sayHello();
36 }
37
38 int main(void)
39 {
40     sayHello();
41     return 0;
42 }
```



Programs versus Software Products



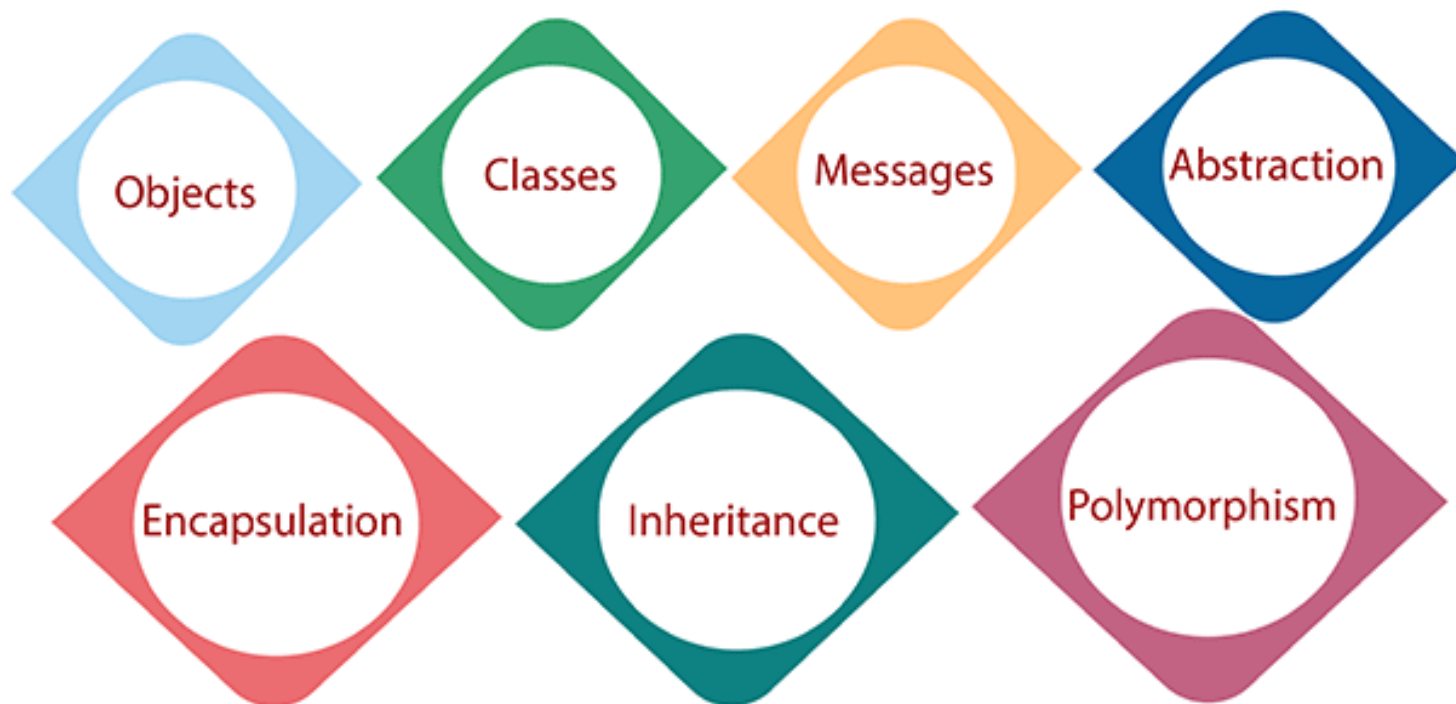
• Usually small in size	• Large
• Author himself is sole user	• Large number of users
• Single developer	• Team of developers
• Lacks proper user interface	• Well-designed interface
• Lacks proper documentation	• Well documented & user-manual prepared
• Ad hoc development.	• Systematic development

Older vs Modern engg methods

Exploratory Style vs. Software Engineering Method

- | | |
|---|--|
| <ul style="list-style-type: none">☐ Based on error correction☐ Finding errors during final product testing☐ Coding is the goal, creating quickly a working system, and modifying till satisfactory | <ul style="list-style-type: none">☐ Focused to error prevention☐ Find errors as early as possible☐ A structured approach, clear specifications, clear phases☐ Periodic reviews during all stages of the project |
|---|--|
-

Object Oriented Design



Object-Oriented Design



- Object-oriented technique:
 - natural objects (such as employees, student, books etc.) occurring in a problem are first identified.
- Relationships among objects:
 - such as composition, reference, and inheritance are determined.

Evolution of Other Software Engineering Techniques (NEW)

- life cycle models,
- specification techniques,
- project management techniques,
- testing techniques,
- debugging techniques,
- quality assurance techniques,
- software measurement techniques,
- CASE tools, etc.

Differences between the exploratory style and modern software development practices



- Use of Life Cycle Models
- Software is developed through several well-defined stages:
 - requirements analysis and specification,
 - design,
 - coding,
 - testing, etc.

Differences between the exploratory style and modern software development practices



- Emphasis has shifted
 - from error correction to error prevention.
- Modern practices emphasize:
 - detection of errors as close to their point of introduction as possible.

Differences between the exploratory style and modern software development practices (CONT.)

- In exploratory style,
 - errors are detected only during testing,
- Now,
 - focus is on detecting as many errors as possible in each phase of development.

Differences between the exploratory style and modern software development practices (CONT.)

- During all stages of development process:
 - Periodic reviews are being carried out
- Software testing has become systematic:
 - standard testing techniques are available.

Differences between the exploratory style and modern software development practices (CONT.)

- Projects are being thoroughly planned:
 - estimation,
 - scheduling,
 - monitoring mechanisms.
- Use of CASE tools.

Life Cycle Model

- A software life cycle model (or process model):
 - a descriptive and diagrammatic model of software life cycle:
 - identifies all the activities required for product development,
 - establishes a precedence ordering among the different activities,
 - Divides life cycle into phases.

Some examples of Software companies in India



Just for Your Information.....



- Some of the Software or Websites Developed By **TCS** are-
 - 1-Passport Seva Kendra Portal(Online Passport Portal For India)
- Some of the Software or Websites Developed By **Infosys** are-
 - 1-Finacle (A banking Software)
 - 2-Airtel Money

Find out net worth of:

- Infosys
- IBM
- TCS
- Tech Mahindra
- Amazon
- Microsoft



Life Cycle Model



Why Model Life Cycle ?



- A written description:
 - forms a common understanding of activities among the software developers.
 - helps in identifying inconsistencies, redundancies in the development process.

Why Model Life Cycle ?



- Processes are tailored for special projects.
 - A documented process model
 - * helps to identify where the tailoring is to occur.

Life Cycle Model (CONT.)



- The development team must identify a suitable life cycle model:
 - and then adhere to it.
 - Primary advantage of adhering to a life cycle model:
 - * helps development of software in a systematic and disciplined manner.

Life Cycle Model (CONT.)



- When a software product is being developed by a team:
 - there must be a precise understanding among team members as to when to do what,
 - otherwise it would lead to and project failure.

Life Cycle Model (CONT.)



- A life cycle model:
 - defines entry and exit criteria for every phase.
 - A phase is considered to be complete:
 - * only when all its exit criteria's are satisfied.

Life Cycle Model (CONT.)



- The phase exit criteria for the software requirements specification phase:
 - Software Requirements Specification (SRS) document is complete, reviewed, and approved by the customer.
- A phase can start:
 - only if its phase-entry criteria have been satisfied.

Life Cycle Model (CONT.)



- It becomes easier for software project managers:
 - to monitor the progress of the project.

Life Cycle Model (CONT.)



- Many life cycle models have been proposed.
- We will confine our attention to a few important and commonly used models.
 - classical waterfall model
 - iterative waterfall,
 - evolutionary,
 - prototyping, and
 - spiral model

SE vs DEVOPS



- DevOps isn't one size fits all

